NATHANAEL MUTUA

Software Developer | React & Full-Stack Engineering | API Design & Integration

Nairobi, Kenya | +254-718-796-084 | nathanael.mutua.m@gmail.com

linkedin.com/in/nxthxnael | github.com/nxthxnael

PROFILE

Final-year Software Engineering student at Murang'a University of Technology (graduating Aug 2026), based in Nairobi, with strong hands-on experience in full-stack web development using React on the frontend and Node.js/Go on the backend. Proven ability to build, consume, and maintain RESTful APIs; design and optimise relational database schemas; debug and troubleshoot production systems; and collaborate effectively in Agile sprint teams. Familiar with object-oriented programming principles, data structures, and software development fundamentals across multiple languages. Eager to apply these skills within any environment and committed to contribute to the existing stacks. Motivated by the opportunity to build software that directly supports financial programmes and communities across Africa.

PROFESSIONAL EXPERIENCE

Software Engineer Attaché · Teach2Give (The Jitu) · Nyeri, Kenya *May – Aug 2025*

- Designed and developed 2 production-grade web applications end-to-end — building React frontend UIs, Node.js backend services, and RESTful APIs with optimised relational database schemas, following existing design systems and component conventions throughout.
- Built, consumed, and maintained 9 RESTful APIs handling authentication, business logic, and data delivery from multiple backend sources, maintaining clean API contracts and structured documentation for frontend integration.
- Implemented UI components, data views, and form interfaces in React following established component library patterns, collaborating closely with the team to ensure consistent, scalable frontend architecture.
- Debugged and troubleshot frontend and backend issues systematically, tracing errors through the API, database, and UI layers to identify root causes and resolve them with minimal disruption to production systems.
- Wrote and maintained technical documentation covering API specifications, architecture decisions, and setup procedures, supporting knowledge transfer and long-term maintainability of the codebase.
- Managed cloud deployments via CI/CD pipelines (Vercel, Render, Railway) using Git-based workflows, following team branching and commit conventions to ensure safe, reproducible releases.
- Implemented OAuth2-based authentication and session security across distributed backend services, applying object-oriented design principles and SQL query optimisation throughout.

Flutter Mobile Developer Intern · Excelerate | Saint Louis University · Remote *Jan – Feb 2026*

- Built 12 reusable UI components for a cross-platform learning application, implementing interfaces following existing design system guidelines with 100% PR coverage, contributing to code reviews that improved quality and prevented regressions across 6+ sprint cycles.
- Integrated REST APIs into user-facing features, handling data fetching, authentication flows, error boundaries, and loading states to deliver a reliable, well-tested user experience.
- Participated in stand-ups, sprint planning, and technical discussions within a 4-developer Agile team, translating business and programme requirements into concrete technical solutions and delivering 100% of features on time.
- Collaborated with backend engineers to design API contracts that supported frontend data requirements cleanly, documenting integration decisions and flagging edge cases to senior developers for resolution.

Project Management Associate Intern · Excelerate | Saint Louis University · Remote*May – Jun 2026*

- Orchestrated cross-functional Jira workflows across 4 concurrent project lifecycles, achieving 100% milestone compliance and a 20% reduction in design-to-delivery timelines through disciplined sprint planning and issue tracking.
- Served as a communication link between technical and non-technical stakeholders, translating business requirements into development tasks and maintaining clear written documentation of decisions, blockers, and resolutions.

- Worked independently across time zones in a fully remote Agile environment, demonstrating the self-management and async communication discipline needed for hybrid distributed team settings.

NOTABLE PROJECTS

Full-Stack E-Commerce Application — *React, TypeScript, Node.js, Drizzle ORM, PostgreSQL, REST API*

- Designed and developed a full-stack web application end-to-end, React frontend with reusable component architecture, Node.js backend, type-safe Drizzle ORM schema across 4 relational data models, and authenticated REST APIs, reducing runtime errors by 40% through rigorous TypeScript discipline and structured code review.
- Applied object-oriented programming principles and relational database design best practices throughout, optimising SQL queries and data model relationships for performance and scalability.

BizSawa / FinAI (GITEX 2026 Contender) — *React Native, Node.js, TypeScript, PostgreSQL, REST APIs, LLM Integration*

- Built a production mobile application with a full backend integration layer, REST API design, authentication, data delivery across 12+ screens, and LLM agent orchestration, demonstrating end-to-end ownership across frontend, backend, and API layers.
- Wrote and executed integration tests for the API layer, catching edge-case failures before production and maintaining documented test coverage aligned with quality assurance standards.

UI Agent Chrome Extension (Final Year Project) — *TypeScript, React, Chrome Extensions API, LLM APIs, Node.js data pipeline*

- Developed a full-stack browser tool end-to-end, designing the data pipeline (input ingestion → API integration → structured output parsing → export) and building the React UI layer, demonstrating ability to contribute across the full software development lifecycle.
- Documented edge cases, technical decisions, and integration patterns throughout, maintaining a clear knowledge base that supported iterative development and potential handoff to other contributors.

Mobile Learning Platform — *Flutter, Dart, REST API integration, SQL-backed offline sync*

- Built a mobile application managing ~56 discrete data points with 75% offline persistence, implementing resilient API sync and local caching strategies, gaining practical experience with data reliability patterns applicable to field-facing digital finance tools.

EDUCATION

B.Sc. Software Engineering — Murang'a University of Technology, Nairobi *2022 – Aug 2026*

Relevant coursework: Software Architecture, Data Structures & Algorithms, Database Systems, Object-Oriented Programming, Distributed Systems, Software Quality Assurance, Human-Computer Interaction

TECHNICAL SKILLS

Frontend: React.js, React Native (Expo), Flutter, TypeScript, JavaScript (ES6+), HTML5, CSS3, responsive design, component-based UI architecture, Visual Studio Code

Backend & APIs: Node.js, Express, Go (Fiber, Gin), RESTful API design & integration, OAuth2/JWT authentication, OOP principles, scalable service architecture

Databases: PostgreSQL, MongoDB, Prisma, Drizzle ORM, SQL schema design, relational data modelling, query optimisation; familiar with Microsoft SQL Server concepts

Testing & QA: Unit & integration test writing, API testing (Postman, Bruno), PR-based code review, debugging & troubleshooting, regression prevention

Dev Tools & Process: Git/GitHub (branching, PRs, commit conventions), Jira, Agile/Scrum, CI/CD pipelines, Docker, Vercel, Render, Railway

Learning / Adjacent: C# / .NET Core fundamentals (actively learning), Python (pandas, numpy, scikit-learn), data pipeline design

Communication: Technical documentation, cross-functional written communication, knowledge base maintenance, async remote collaboration